mhs-poisson-z: a pressure-only magnetohydrostatic correction

The physics of `run.mode = "mhs-poisson-z"`, explained step by step (B.Sc. level). HSEMHS.

1. The question

The mode `column-hse-z` puts every vertical column into hydrostatic equilibrium on its own. Neighbouring columns then push on each other with forces nothing balances, and the magnetic field is ignored. Here we keep the temperature T and the magnetic field $\mathbf{B}$ exactly as given and ask: *which pressure field balances, as well as possible, gravity and the magnetic force in all three directions?* Only the pressure changes. This idea comes from the FIRTEZ-dz inversion code (Borrero et al.): inside an inversion the atmosphere changes at every iteration, and a pressure-only correction can restore the force balance in minutes, whereas changing the field as well takes far longer.

2. The quantities

Symbol	Meaning	Unit
x, y, z	horizontal coordinates (the box is periodic in x and y) and height	m
T, $\mathbf{B}$	temperature and magnetic field, given, **both fixed**	K, T
$\mathbf{J}$	electric current density, $\mathbf{J} = \nabla \times \mathbf{B}/\mu_0$	$\mathrm{A\,m^{-2}}$
$\mathbf{L}$	Lorentz force per volume, $\mathbf{L} = \mathbf{J} \times \mathbf{B}$; fixed, because $\mathbf{B}$ is fixed	$\mathrm{N\,m^{-3}}$
P, u	gas pressure, the only unknown; $u = \ln P$	Pa, –
ρ	density, $\rho(T, P)$ from the equation of state	$\mathrm{kg\,m^{-3}}$
g	gravity, 274; the gravitational force points along $-\hat{\mathbf{z}}$	$\mathrm{m\,s^{-2}}$
H, $\bar{H}$	pressure scale height $P/(\rho g)$; its median over a horizontal layer	m
$\mathbf{F}$	net force per volume on the gas	$\mathrm{N\,m^{-3}}$
β	plasma beta, $2\mu_0 P/B^2$: gas pressure over magnetic pressure	–
μ_0	vacuum permeability, $4\pi \times 10^{-7}$	$\mathrm{T\,m\,A^{-1}}$
z_{cut}	top of the region in which the force balance is solved	m

3. The forces

A small volume of gas feels three forces per unit volume:

1. **pressure**, $-\nabla P$: gas is pushed from high to low pressure;
2. **gravity**, $-\rho g\,\hat{\mathbf{z}}$;
3. the **Lorentz force** $\mathbf{L} = \mathbf{J} \times \mathbf{B}$: the field acts on the electric currents flowing in the ionised gas.

The currents follow from the field by Ampère's law (in a slowly changing plasma), $\mathbf{J} = \nabla \times \mathbf{B}/\mu_0$. With a vector identity the Lorentz force splits into two parts with a clear meaning:

$$\mathbf{J} \times \mathbf{B} = \frac{(\nabla \times \mathbf{B}) \times \mathbf{B}}{\mu_0} = \underbrace{-\nabla\left(\frac{B^2}{2\mu_0}\right)}_{\text{magnetic pressure}} + \underbrace{\frac{(\mathbf{B} \cdot \nabla)\mathbf{B}}{\mu_0}}_{\text{magnetic tension}} .$$

Field lines push sideways on each other like a gas with pressure $B^2/2\mu_0$, and curved field lines pull like stretched rubber bands. The gas is static when the forces cancel:

$$\boxed{\mathbf{F} \equiv -\nabla P + \mathbf{J} \times \mathbf{B} - \rho g\,\hat{\mathbf{z}} = 0}$$

In this mode $\mathbf{J} \times \mathbf{B}$ is a *given* force field (computed once from the fixed $\mathbf{B}$), and ρ depends on P through the equation of state. The question is therefore which pressure balances gravity plus a given force field.

4. Why it is called a Poisson step

Take the divergence of $\mathbf{F} = 0$, using $\nabla \cdot \nabla P = \nabla^2 P$:

$$\nabla^2 P = \nabla \cdot (\mathbf{J} \times \mathbf{B}) - g\,\frac{\partial \rho}{\partial z}.$$

This is a **Poisson equation** for P, the same type of equation as for the electric potential of a charge distribution. FIRTEZ solves essentially this, iterating because ρ depends on P. Poisson equations can be solved very fast in a periodic box (Sect. 9).

5. What a pressure correction can do, and what it cannot

5.1 Gradient and non-gradient forces

A force field in the box can be split (the **Helmholtz decomposition**) into

$$\mathbf{F} = \underbrace{-\nabla\phi}_{\text{gradient part}} + \underbrace{\mathbf{S}}_{\text{non-gradient part}}, \qquad \int \mathbf{S}\cdot\nabla\psi\, dV = 0 \text{ for every function } \psi.$$

The gradient part is the slope of a scalar "potential" ϕ, like a pressure force, and has no curl. The non-gradient part is everything else: it carries the whole curl of $\mathbf{F}$, and the orthogonality condition means (after integrating by parts) that it has no divergence. The two parts are orthogonal, so their sizes add like the sides of a right triangle: $\int |\mathbf{F}|^2 dV = \int |\nabla\phi|^2 dV + \int |\mathbf{S}|^2 dV$. The split is unique only once boundary conditions are fixed: our box is periodic in x and y but has a top and a bottom, so what happens at those boundaries (and an additive constant in ϕ) has to be specified. With a weighting of the volume, as below, "orthogonal" is meant in that weighted sense.

A change of pressure δP changes the force by $-\nabla\delta P$, a pure gradient. If the density did not change, a pressure correction could in principle remove the gradient part (choose $\delta P = -\phi$) and could never touch $\mathbf{S}$. This picture explains what the method can achieve; the solver itself does not compute such a projection. It minimises a weighted sum of squares of the *relative* residual (Sect. 7), which is a non-linear problem, so it neither removes the gradient part exactly nor leaves $\mathbf{S}$ untouched.

5.2 Which forces are non-gradient?

Take the curl of the force balance. The curl of any gradient is zero, and for gravity $\nabla\times(\rho g\hat{\mathbf{z}}) = g\,(\partial_y\rho,\, -\partial_x\rho,\, 0)$, so

$$\nabla\times\mathbf{F} = \nabla\times(\mathbf{J}\times\mathbf{B}) - g\left(\frac{\partial\rho}{\partial y},\, -\frac{\partial\rho}{\partial x},\, 0\right).$$

Term by term:

- **pressure**: always a gradient;
- **gravity**: a gradient only if ρ is the same everywhere on each horizontal plane. A heavy column next to a light one at the same height cannot be static: the weight of the heavy column makes the pressure below it rise faster with depth, and the resulting sideways push has nowhere to go. This is **buoyancy**;
- **Lorentz force**: the magnetic pressure is a gradient, the tension in general is not.

So static equilibrium requires the **curl condition**

$$\nabla\times(\mathbf{J}\times\mathbf{B}) = g\left(\frac{\partial\rho}{\partial y},\, -\frac{\partial\rho}{\partial x},\, 0\right):$$

only the tension of the field can hold up horizontal density differences. Without a field this forces $\partial_x\rho = \partial_y\rho = 0$: a static non-magnetic atmosphere is horizontally uniform. With a given $\mathbf{B}$ the curl condition holds only if the field happens to have exactly the right tension everywhere. There is no reason to expect this of a field taken from a moving simulation or an inversion, so in general *no pressure makes the atmosphere exactly static*; for a particular field this has to be checked, not assumed. The mode looks for the best compromise.

5.3 The density loophole

The density is not fixed: with T fixed, a pressure change changes it,

$$\delta\rho = \rho\,\frac{\partial\ln\rho}{\partial\ln P}\bigg|_T\frac{\delta P}{P}, \qquad \text{so} \qquad \delta(\nabla\times\mathbf{F}) = -g\,\nabla\times(\delta\rho\,\hat{\mathbf{z}}).$$

By making the pressure fall faster with height under a dense column, the correction lowers that column's density at a given height and shrinks the horizontal density contrast, and with it part of $\mathbf{S}$. What it can never remove is the part caused by the temperature differences at fixed pressure, or by the tension of $\mathbf{B}$: only a change of $\mathbf{B}$ (the `full-mhs-z` mode) or of T can do that.

6. Measuring the imbalance in units of the weight

Divide the force by the pressure and use $u = \ln P$ (so $\nabla P/P = \nabla u$ and $\rho g/P = 1/H$):

$$\boxed{\mathbf{R} \equiv \frac{\mathbf{F}}{P} = -\nabla u - \frac{\hat{\mathbf{z}}}{H(T,P)} + \frac{\mathbf{J}\times\mathbf{B}}{P}}$$

1. $P = e^u$ is positive automatically, across the many decades of pressure in the box.

2. **R** is a *relative* imbalance. If we minimised $|\mathbf{F}|$ itself, the cheapest way to make the force small would be to make the pressure small everywhere (all forces scale with P); dividing by P removes that false incentive.
3. Without the Lorentz force, $R_z = 0$ is $du/dz = -1/H$, ordinary hydrostatic equilibrium.
4. $R\,\bar{H} \approx F/(\rho g)$: multiplied by the scale height, the residual is the unbalanced force in units of the weight of the gas. A value of 1 means an unbalanced force as large as gravity.

7. Least squares

At every point the force balance gives three equations (x, y, z), but there is only one unknown, the pressure. With more equations than unknowns and no exact solution, we look for the **best compromise**: the pressure that makes the sum of the squared residuals as small as possible, just as fitting a straight line through many points minimises the squared distances. The quantity minimised is

$$\Phi(u) = \sum_{\text{points}} w\,\bar{H}^2\left(R_x^2 + R_y^2\right) + \sum_{\text{between layers}} h\,\bar{H}^2 R_z^2.$$

w and h are the thicknesses of the cells, so Φ approximates an integral over the volume. $\bar{H}$ (the layer median of H in the starting state) makes each term a force in units of the local weight, so the deep, dense layers, whose forces are many orders of magnitude larger, do not swamp the upper ones. Force balance hardly fixes the overall level of the pressure: for an ideal gas of fixed ionisation, multiplying P everywhere by a constant leaves ∇u and $1/H$ unchanged, so only the Lorentz term $\mathbf{J}\times\mathbf{B}/P$ (and the weak pressure dependence of the ionisation) would notice. The level is therefore fixed separately: each step is penalised for changing the mean of u on the bottom plane, which keeps that mean close to its starting (column-HSE) value. It is a penalty, not an exact constraint, and the limit on the size of each step can shift it slightly.

8. The low-β problem and the cut at z_{cut}

The ratio $\beta = 2\mu_0 P/B^2$ compares gas and magnetic pressure. Deep down $\beta \gg 1$: the gas dominates. Higher up $\beta \ll 1$: there the Lorentz force dwarfs every gas force, and a static state needs a nearly **force-free** field, $\mathbf{J}\times\mathbf{B} \approx 0$ (currents flowing along the field lines). A field from a dynamic simulation or an inversion is not force-free. With **B** fixed, the only way to balance a large Lorentz force is a large pressure gradient, i.e. a large pressure. And because $\partial P/\partial z$ links each layer to the one below, raising the pressure high up raises it in the photosphere too, which would destroy the result where it matters most.

The force balance is therefore solved only below a height z_{cut} (by default 1 Mm), where the gas still dominates. Above it each column is integrated in plain hydrostatic equilibrium, $du/dz = -1/H$, starting from the solved pressure at z_{cut}. FIRTEZ makes the same physical choice more gradually (its magnetic term decays exponentially above a chosen height). z_{cut} is a physical parameter: it marks where we stop trusting the given field to be in equilibrium. Above it the atmosphere is hydrostatic, not magnetohydrostatic.

9. How the minimum is found

9.1 Gauss–Newton

R depends on u linearly through $-\nabla u$, but non-linearly through $\rho(T, P)$ and the factor $1/P$ in the Lorentz term. In one dimension, Newton's method finds a root of $f(u)$ by replacing f with its tangent and solving $f(u)+f'(u)\,\delta u = 0$. The many-variable least-squares version, **Gauss–Newton**, replaces **R** by its linear approximation $\mathbf{R}(u+\delta u) \approx \mathbf{R}(u) + \mathbf{J}_R\,\delta u$ ($\mathbf{J}_R$ the matrix of derivatives) and minimises the resulting quadratic in δu. Setting its derivative to zero gives the **normal equations**

$$\mathbf{J}_R^\top W \mathbf{J}_R\,\delta u = -\mathbf{J}_R^\top W \mathbf{R},$$

with W the weights of Φ. Since $\mathbf{J}_R$ contains the derivatives $-\nabla$, the leading part of $\mathbf{J}_R^\top W \mathbf{J}_R$ is $-\nabla\cdot(W\nabla\,\cdot)$, a weighted Laplacian: *each step is a Poisson equation for* δu, plus local terms from the equation of state and the Lorentz force. The step is limited (the pressure may change by at most a factor e^2 per point and step) and halved until Φ actually decreases. The iteration stops when Φ has stopped decreasing.

9.2 Fourier modes

In a horizontally periodic box any function can be written as a sum of waves, $u(x,y,z) = \sum_{k_x,k_y} \hat{u}(k_x,k_y,z)\,e^{i(k_x x + k_y y)}$. A horizontal derivative acts on a wave as a multiplication, $\partial_x \to ik_x$, so the Laplacian becomes $-k_x^2 - k_y^2 + \partial_z^2$ and each wave (k_x, k_y) obeys its own one-dimensional equation in z. On the grid each is a tridiagonal linear system, solved in a number of operations proportional to the number of layers. This is the fast approximate solver (the "preconditioner") that guides the iterative solution of each Gauss–Newton step.

10. Optical depth

With P known everywhere, the equation of state gives ρ and the 500 nm extinction χ_{500}, and

$$\tau_{500}(z) = \tau_{\text{top}} + \int_z^{z_{\text{top}}} \chi_{500}\, dz',$$

with the gas above the box approximated by an exponentially decreasing χ (as in `column-hse-z`). The height of $\tau_{500} = 1$ is the visible surface; since the opacity grows with density, any change of pressure moves it.

11. Summary

The mode reduces, as far as a pressure change can, the force imbalance left by independent hydrostatic columns: mostly the gradient part, and through the density part of the non-gradient part. It minimises a residual; it does not guarantee an exact equilibrium. The rest, the buoyancy of the given temperature structure that the given field cannot hold up, requires the field to change. Above z_{cut} the atmosphere is only hydrostatic.